Proof

Steps 2–3: from the discrete objective to the continuum integral, the HJB equation, and the matrix Riccati ODE

Full derivation — see WorkPlan.tex Stage 2 for the condensed version and Summary.tex for the one-li

Notation

Symbol	Meaning
M	Number of assets.
N, τ	Number of discrete time steps, and the step size ($\tau = T/N$), in Step 1's discretisation.
j, k	Asset index ($j = 1, \dots, M$) and time-step index ($k = 1, \dots, N$).
γ_j	Permanent-impact coefficient for asset j.
ε_j	Spread-cost coefficient for asset j.
$\tilde{\eta}_j$	Temporary-impact coefficient for asset j; $H = \mathrm{diag}(\tilde{\eta}_j)$.
n_{jk}	Shares of asset j traded at step k (discrete).
$\mathbf{x}_k$, $\mathbf{x}(t)$	Inventory vector at step k (discrete) / time t (continuous).
$\mathbf{v}(t)$	Selling-rate vector, $\mathbf{v}(t) = -\dot{\mathbf{x}}(t)$.
$\mathbf{X}$	Initial inventory vector, $\mathbf{x}(0) = \mathbf{X}$.
T	Liquidation horizon.
λ	Risk-aversion parameter.
C	Risk/covariance matrix ($M \times M$, symmetric, positive semi-definite).
H	Temporary-impact matrix ($M \times M$, diagonal, positive-definite), $H = \mathrm{diag}(\tilde{\eta}_j)$.
$J[\mathbf{v}]$	The continuum objective, (2), to be minimised over schedules $\mathbf{v}(\cdot)$.
Φ	Terminal penalty matrix; $\mathcal{V}(\mathbf{x}, T) = \frac{1}{2}\mathbf{x}^\top \Phi \mathbf{x}$ enforces $\mathbf{x}(T) = \mathbf{0}$ as $\Phi \to \infty$.
$\mathcal{V}(\mathbf{x}, t)$	Cost-to-go (value function) from state $\mathbf{x}$ at time t.
$P(t)$	Symmetric matrix with $\mathcal{V}(\mathbf{x}, t) = \frac{1}{2}\mathbf{x}^\top P(t)\mathbf{x}$; solves the Riccati ODE (5).
$\mathbf{v}^*$	Optimal feedback control, $\mathbf{v}^* = \frac{1}{2}H^{-1}P(t)\mathbf{x} = \frac{1}{2}H^{-1}\nabla_{\mathbf{x}}\mathcal{V}$.
U, μ_i	Orthogonal eigenbasis and eigenvalues of $H^{-1/2}CH^{-1/2} = U\,\mathrm{diag}(\mu_i)\,U^\top$.
κ_i	Decay rate of decoupled mode i, $\kappa_i = \sqrt{\lambda\mu_i}$.
$Q(t)$, $\hat{Q}(t)$, $q_i(t)$	Rescaled ($H^{-1/2}PH^{-1/2}$), rotated ($U^\top QU$), and diagonal-entry forms of $P(t)$ used in the Corollary's proof.
$\phi(t)$	Riccati-linearising substitution, $q = -2\dot{\phi}/\phi$, used to solve the scalar Riccati ODE in closed form.
$\mathbf{z}(t)$, $z_i(t)$, K_i	Decoupled coordinates $\mathbf{z} = U^\top H^{1/2}\mathbf{x}$, their components, and the integration constants fixing $z_i(0)$.

Step 2 — from the discrete objective to the continuum integral

Starting point, Step 1's discrete objective (Summary.tex Step 1):

$$U[\mathbf{x}] = \sum_{j=1}^{M}\left[\tfrac{1}{2}\gamma_j X_j^2 + \varepsilon_j \sum_{k=1}^{N}|n_{jk}| + \frac{\tilde{\eta}_j}{\tau}\sum_{k=1}^{N} n_{jk}^2\right] + \lambda \sum_{k=1}^{N} \tau\, \mathbf{x}_k^\top C \mathbf{x}_k. \tag{1}$$

1. Drop the two schedule-independent terms. $\frac{1}{2}\gamma_j X_j^2$ (permanent impact) depends only on the total amount liquidated, not on how it is spread over time, so it is a constant over all

schedules. Under the sales-only assumption A7 ($n_{jk} \geq 0$ always), $|n_{jk}| = n_{jk}$ and $\sum_k n_{jk} = X_j$, so $\varepsilon_j \sum_k |n_{jk}| = \varepsilon_j X_j$ is also a constant. Neither term can be changed by choosing a different schedule, so both drop out of the optimization, leaving

$$\tilde{U}[\mathbf{x}] = \sum_j \frac{\tilde{\eta}_j}{\tau} \sum_k n_{jk}^2 + \lambda \sum_k \tau\, \mathbf{x}_k^\top C \mathbf{x}_k.$$

2. Substitute $n_{jk} = \tau v_j(t_k)$ (trade size = rate × time-step):

$$\frac{\tilde{\eta}_j}{\tau} \sum_k n_{jk}^2 = \frac{\tilde{\eta}_j}{\tau} \sum_k \tau^2 v_j(t_k)^2 = \sum_k \tau\, \tilde{\eta}_j\, v_j(t_k)^2.$$

Stacking over j with $H = \mathrm{diag}(\tilde{\eta}_j)$:

$$\tilde{U} = \sum_k \tau \Big(\mathbf{v}(t_k)^\top H\, \mathbf{v}(t_k) + \lambda\, \mathbf{x}(t_k)^\top C\, \mathbf{x}(t_k) \Big),$$

a Riemann sum with step size τ.

3. Take the continuum limit $\tau \to 0$ ($N = T/\tau \to \infty$). A Riemann sum $\sum_k \tau f(t_k) \to \int_0^T f(t)\, dt$ for bounded, (piecewise) continuous f, so

$$J[\mathbf{v}] = \int_0^T \big(\mathbf{v}(t)^\top H \mathbf{v}(t) + \lambda\, \mathbf{x}(t)^\top C \mathbf{x}(t) \big)\, dt. \tag{2}$$

4. The inventory recursion becomes the state equation. $\mathbf{x}_k = \mathbf{x}_{k-1} - \mathbf{n}_k = \mathbf{x}_{k-1} - \tau\, \mathbf{v}(t_k)$, so $(\mathbf{x}_k - \mathbf{x}_{k-1})/\tau = -\mathbf{v}(t_k)$; as $\tau \to 0$ this is the definition of the derivative,

$$\dot{\mathbf{x}} = -\mathbf{v}, \qquad \mathbf{x}(0) = \mathbf{X}, \qquad \mathbf{x}(T) = \mathbf{0}, \tag{3}$$

the boundary conditions carrying over directly from the discrete recursion's endpoints.

This is the multi-asset continuum limit of Bergault–Drissi–Guéant [2], Sec. 2 and App. A; the single-asset case matches Almgren–Chriss [1], Sec. 2.2 and App. A.2.

Step 3 — what is being proved next

Starting from Step 2's continuum objective (2)–(3), with H diagonal positive-definite and C symmetric positive semi-definite (assumptions A2/A11, A13), this note derives, by Bellman's dynamic programming principle (not Euler–Lagrange, since a state-space PDE is what Stage 7's Schrödingerisation needs downstream), exactly the two boxed results of Summary.tex Step 3:

$$\partial_t \mathcal{V} - \tfrac{1}{4} (\nabla_{\mathbf{x}} \mathcal{V})^\top H^{-1} (\nabla_{\mathbf{x}} \mathcal{V}) + \lambda\, \mathbf{x}^\top C \mathbf{x} = 0, \tag{4}$$

$$\dot{P} = \tfrac{1}{2} P H^{-1} P - 2\lambda C, \qquad P(T) = \Phi. \tag{5}$$

A corollary at the end also recovers the closed-form trajectory used in Stage 2–3, for the hard-liquidation limit $\Phi \to \infty$.

Step A — Bellman's principle gives the HJB PDE

Define the cost-to-go from state $\mathbf{x}$ at time t:

$$\mathcal{V}(\mathbf{x}, t) = \min_{\mathbf{v}(\cdot)} \int_t^T \big(\mathbf{v}(s)^\top H \mathbf{v}(s) + \lambda\, \mathbf{x}(s)^\top C \mathbf{x}(s) \big)\, ds, \qquad \dot{\mathbf{x}} = -\mathbf{v}, \quad \mathbf{x}(t) = \mathbf{x}.$$

Split the horizon into $[t, t+dt]$ and $[t+dt, T]$ and use that the second piece, started from whatever state the first piece reaches, is again optimal (Bellman's principle):

$$\mathcal{V}(\mathbf{x}, t) = \min_{\mathbf{v}} \Big[\big(\mathbf{v}^\top H \mathbf{v} + \lambda\, \mathbf{x}^\top C \mathbf{x} \big)\, dt + \mathcal{V}(\mathbf{x} - \mathbf{v}\, dt,\, t + dt) \Big] + o(dt).$$

Taylor-expand the second term to first order in dt:

$$\mathcal{V}(\mathbf{x} - \mathbf{v}\, dt,\, t + dt) = \mathcal{V}(\mathbf{x}, t) + \partial_t \mathcal{V}\, dt - \mathbf{v}^\top \nabla_{\mathbf{x}} \mathcal{V}\, dt + o(dt).$$

Substitute back, cancel $\mathcal{V}(\mathbf{x}, t)$ from both sides, divide by dt, and let $dt \to 0$:

$$0 = \min_{\mathbf{v}} \Big[\mathbf{v}^\top H \mathbf{v} - \mathbf{v}^\top \nabla_{\mathbf{x}} \mathcal{V} \Big] + \partial_t \mathcal{V} + \lambda\, \mathbf{x}^\top C \mathbf{x}. \tag{6}$$

Step B — pointwise minimization: the optimal feedback control

The bracket in (6) is a convex quadratic in $\mathbf{v}$ (since $H \succ 0$). Setting its gradient to zero,

$$2H\mathbf{v} - \nabla_{\mathbf{x}} \mathcal{V} = 0 \quad \Longrightarrow \quad \mathbf{v}^* = \tfrac{1}{2} H^{-1} \nabla_{\mathbf{x}} \mathcal{V}.$$

Substituting $\mathbf{v}^*$ back into the bracket:

$$\mathbf{v}^{*\top} H \mathbf{v}^* - \mathbf{v}^{*\top} \nabla_{\mathbf{x}} \mathcal{V} = \tfrac{1}{4} (\nabla_{\mathbf{x}} \mathcal{V})^\top H^{-1} (\nabla_{\mathbf{x}} \mathcal{V}) - \tfrac{1}{2} (\nabla_{\mathbf{x}} \mathcal{V})^\top H^{-1} (\nabla_{\mathbf{x}} \mathcal{V}) = -\tfrac{1}{4} (\nabla_{\mathbf{x}} \mathcal{V})^\top H^{-1} (\nabla_{\mathbf{x}} \mathcal{V}).$$

Plugging this into (6) gives exactly (4), with terminal condition $\mathcal{V}(\mathbf{x}, T) = \frac{1}{2} \mathbf{x}^\top \Phi \mathbf{x}$ (the penalty form of the hard constraint $\mathbf{x}(T) = \mathbf{0}$, A10). This matches Almgren–Chriss [1], App. A.2, and Bergault–Drissi–Guéant [2], Prop. 1 (their Eqs. 13–14, under the identification $\eta = H$, $\Sigma = C$, $\gamma = 2\lambda$, no Ornstein–Uhlenbeck signal, i.e. $R = 0$).

Step C — the quadratic ansatz collapses the PDE to a Riccati ODE

Try $\mathcal{V}(\mathbf{x}, t) = \frac{1}{2} \mathbf{x}^\top P(t) \mathbf{x}$ with $P(t)$ symmetric. Then $\partial_t \mathcal{V} = \frac{1}{2} \mathbf{x}^\top \dot{P} \mathbf{x}$ and $\nabla_{\mathbf{x}} \mathcal{V} = P(t) \mathbf{x}$. Substituting into (4):

$$\tfrac{1}{2} \mathbf{x}^\top \dot{P} \mathbf{x} - \tfrac{1}{4} \mathbf{x}^\top P H^{-1} P \mathbf{x} + \lambda\, \mathbf{x}^\top C \mathbf{x} = 0 \qquad \text{for all } \mathbf{x}.$$

A quadratic form $\mathbf{x}^\top M \mathbf{x}$ vanishes for all $\mathbf{x}$ iff the symmetric part of M is zero; since $\dot{P}$, $PH^{-1}P$ and C are each already symmetric, this forces

$$\tfrac{1}{2} \dot{P} - \tfrac{1}{4} P H^{-1} P + \lambda C = 0 \quad \Longrightarrow \quad \dot{P} = \tfrac{1}{2} P H^{-1} P - 2\lambda C,$$

which is exactly (5), with $P(T) = \Phi$ read off the terminal condition above. The optimal control becomes the linear feedback law $\mathbf{v}^*(\mathbf{x}, t) = \frac{1}{2} H^{-1} P(t) \mathbf{x}$. This matches Bergault–Drissi–Guéant [2], Prop. 2 and App. A (their Eqs. 20–21, same identification as above, plus $\Gamma = \frac{1}{2}\Phi$, $P_{\text{here}} = -2A_{\text{theirs}}$); the one-asset scalar case matches Almgren–Chriss [1], Eq. 17.

Corollary — closed-form trajectory, $\Phi \to \infty$

Claim. *Let $H^{-1/2} C H^{-1/2} = U \operatorname{diag}(\mu_i) U^\top$ (U orthogonal, $\mu_i \geq 0$ since $C \succeq 0$) and $\kappa_i = \sqrt{\lambda \mu_i}$. As $\Phi \to \infty$ (hard liquidation, $\mathbf{x}(T) = \mathbf{0}$),*

$$\mathbf{x}(t) = H^{-1/2} U \operatorname{diag}\Big(\frac{\sinh \kappa_i (T - t)}{\sinh \kappa_i T} \Big) U^\top H^{1/2} \mathbf{X}.$$

Proof. Write $Q(t) = H^{-1/2}P(t)H^{-1/2}$ (symmetric). Using $H^{-1} = H^{-1/2}H^{-1/2}$,

$$\dot{Q} = H^{-1/2}\dot{P}H^{-1/2} = \tfrac{1}{2}\big(H^{-1/2}PH^{-1/2}\big)\big(H^{-1/2}PH^{-1/2}\big) - 2\lambda H^{-1/2}CH^{-1/2} = \tfrac{1}{2}Q^2 - 2\lambda\tilde{C},$$

with $\tilde{C} = H^{-1/2}CH^{-1/2} = U\,\mathrm{diag}(\mu_i)\,U^\top$. Let $\hat{Q} = U^\top QU$; since U is constant,

$$\dot{\hat{Q}} = \tfrac{1}{2}\hat{Q}^2 - 2\lambda\,\mathrm{diag}(\mu_i).$$

As $\Phi \to \infty$, $Q(T) \to \infty\, I$, so $\hat{Q}(T) \to \infty\, I$, which is diagonal. The diagonal matrices are an invariant subspace of this ODE (a diagonal $\hat{Q}$ keeps $\hat{Q}^2$, hence $\dot{\hat{Q}}$, diagonal), so by uniqueness $\hat{Q}(t) = \mathrm{diag}(q_i(t))$ for all $t \le T$, with each q_i solving the decoupled scalar Riccati equation $\dot{q}_i = \frac{1}{2}q_i^2 - 2\lambda\mu_i$, $q_i(T) \to \infty$.

For a scalar Riccati $\dot{q} = \frac{1}{2}q^2 - 2\lambda\mu$, substitute $q = -2\dot{\phi}/\phi$ (the standard Riccati-to-linear-ODE substitution):

$$\dot{q} = -2\Big(\frac{\ddot{\phi}}{\phi} - \frac{\dot{\phi}^2}{\phi^2}\Big), \qquad \tfrac{1}{2}q^2 = 2\frac{\dot{\phi}^2}{\phi^2},$$

so the Riccati equation becomes $-2\ddot{\phi}/\phi = -2\lambda\mu$, i.e. $\ddot{\phi} = \lambda\mu\,\phi = \kappa_i^2\phi$ with $\kappa_i = \sqrt{\lambda\mu_i}$. The blow-up condition $q_i(T) \to \infty$ means $\phi(T) = 0$; take $\phi(t) = \sinh(\kappa_i(T-t))$ (solves $\ddot{\phi} = \kappa_i^2\phi$ and vanishes at $t = T$), giving $q_i(t) = 2\kappa_i \coth(\kappa_i(T-t))$.

In the decoupled coordinates $\mathbf{z}(t) = U^\top H^{1/2}\mathbf{x}(t)$, the feedback law $\mathbf{v}^* = \frac{1}{2}H^{-1}P\mathbf{x} = H^{-1/2}U\,\mathrm{diag}(\frac{1}{2}q_i)\,U^\top H^{1/}$ together with $\dot{\mathbf{x}} = -\mathbf{v}^*$ gives $\dot{z}_i = -\kappa_i \coth(\kappa_i(T-t))\,z_i$. Integrating, $\ln z_i(t) = \ln\sinh(\kappa_i(T-t)) +$ const, so $z_i(t) = K_i \sinh(\kappa_i(T-t))$; matching $z_i(0) = X_i$ (the i-th component of $U^\top H^{1/2}\mathbf{X}$) fixes $K_i = X_i/\sinh(\kappa_i T)$. Converting back, $\mathbf{x}(t) = H^{-1/2}U\mathbf{z}(t)$, gives the claimed formula. □

For $M = 1$ (scalar $H = \tilde{\eta}$, $C = \sigma^2$) this reduces to $x(t) = X\sinh(\kappa(T-t))/\sinh(\kappa T)$ with $\kappa = \sqrt{\lambda\sigma^2/\tilde{\eta}}$ — Almgren–Chriss [1] Eq. 17, exactly. For finite Φ the same diagonalization argument applies to (5) directly (without the $\Phi \to \infty$ shortcut); see Bergault–Drissi–Guéant [2], App. A, Eqs. 31–34, Cor. 1, and WorkPlan.tex eq. `eq:traj-finite-phi`.

Cross-check against the literature

Every step above is this project's own algebra; the places it is checked against a published source are: (i) Step 2's continuum limit is the multi-asset formulation of Bergault–Drissi–Guéant [2], Sec. 2/App. A, and matches Almgren–Chriss [1], Sec. 2.2/App. A.2, in the single-asset case; (ii) Step B/C's PDE and Riccati ODE match Bergault–Drissi–Guéant [2] Eqs. 13–14 and 20–21 under the stated parameter identification (AI-checked symbol-for-symbol this session); (iii) the one-asset trajectory in the Corollary matches Almgren–Chriss [1] Eq. 17 exactly. None of these checks is a numerical validation — see Summary.tex's status-column legend for what each kind of check does and does not establish.

References

[1] R. Almgren and N. Chriss, "Optimal Execution of Portfolio Transactions," *J. Risk* **3**(2), 5–39 (2000).

[2] P. Bergault, F. Drissi, and O. Guéant, "Multi-asset Optimal Execution and Statistical Arbitrage Strategies under Ornstein–Uhlenbeck Dynamics," *SIAM J. Financial Math.* **13**(1), 353–390 (2022); arXiv:2103.13773.